

#This code will be used to demonstrate theorem 6.18. We will demonstrate that $Q2 = ((w+z)^3 - y^3) \cdot x \cdot (-w^2 \cdot (z^3 - y^3) + z^2 \cdot (w^3 - y^3)) / (y \cdot (w+z)^2 \cdot ((x^3 - w^3) \cdot (z^3 - y^3) + (x^3 - z^3) \cdot (w^3 - y^3)))$ is increasing along the γ_2 curve; (Part 1 – locating potential critical points)

#Define Q2;

$$Q2 := \frac{((w+z)^3 - y^3) \cdot x \cdot (-w^2 \cdot (z^3 - y^3) + z^2 \cdot (w^3 - y^3))}{y \cdot (w+z)^2 \cdot ((x^3 - w^3) \cdot (z^3 - y^3) + (x^3 - z^3) \cdot (w^3 - y^3))} :$$

#Apply the rational parametrization constructed in this chapter 4.

$$\text{parametrization} := \left\{ w = a + \frac{1}{a}, \right.$$

$$z = b + \frac{1}{b},$$

$$x = a - \frac{1}{a} + b - \frac{1}{b},$$

$$y = a - \frac{1}{a} - b + \frac{1}{b} \Big\} :$$

$$q2 := \text{simplify}(\text{expand}(\text{subs}(\text{parametrization}, Q2))) ;$$

$$\begin{aligned} q2 := & \left((a b - 1)^2 (-a^4 b^8 + 2(a^5 - 2a^3) b^7 + (a^6 + 10a^4 - 5a^2) b^6 + 2(-a^7 + 6a^3 \right. & (1) \\ & - 2a) b^5 + (a^8 - 6a^6 + 4a^4 + 10a^2 - 1) b^4 + 2(2a^7 - 6a^5 + a) b^3 + (7a^6 - 6a^4 \\ & + a^2) b^2 + 2(2a^5 - a^3) b + a^4) \left(a^2 + \frac{b^2}{3} \right) b \Bigg) \Bigg/ \left(2 \left(\frac{a^6 b^{12}}{6} + \frac{(a^7 - a^5) b^{11}}{2} \right. \right. \\ & + \frac{(-a^8 - a^4) b^{10}}{2} - \frac{a^3 \left(a^4 + a^2 + \frac{2}{3} \right) b^9}{2} + a^4 \left(a^6 + a^4 + \frac{3}{2} a^2 - 1 \right) b^8 + (-4a^7 \\ & + a^5 - a^3) b^7 + \left(-\frac{1}{3} - a^{10} - 4a^8 + a^4 + a^2 \right) b^6 + (-4a^7 + a^5 - a^3) b^5 + a^4 \left(a^6 + a^4 \right. \\ & + \frac{3}{2} a^2 - 1 \Big) b^4 - \frac{a^3 \left(a^4 + a^2 + \frac{2}{3} \right) b^3}{2} + \frac{(-a^8 - a^4) b^2}{2} + \frac{(a^7 - a^5) b}{2} + \frac{a^6}{6} \Bigg) (a \\ & \left. + b) \right) \end{aligned}$$

#Let us denote by $D1 = \frac{da}{db}$. By the chain rule, the derivative of Q2 is zero if and only if

$$Dl := -\frac{diff(q2,b)}{diff(q2,a)}:$$

#Define r2

$$\begin{aligned} r2 := & \left(-1024 \left(b^2 - \frac{1}{2} \right) b^{13} (b^2 - 5) (b^2 + 1)^4 a^{23} + a^{32} b^{16} + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^2 + \right. \\ & -4 b^{18} + 8 b^{16} - 4 b^{14} \left. \right) a^{30} + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} + b^{16} - 9 \left(b^{16} \right. \\ & + \frac{4168}{9} b^{14} - 528 b^{12} - 328 b^{10} + \frac{3698}{9} b^8 - 72 b^6 - 16 b^4 + 8 b^2 + 1 \left. \right) b^8 a^{24} - 4 b^2 (b^{28} \\ & + 12 b^{26} + 37 b^{24} + 378 b^{22} + 4187 b^{20} + 19656 b^{18} + 41514 b^{16} + 57358 b^{14} + 65514 b^{12} \\ & + 34120 b^{10} + 4315 b^8 + 1018 b^6 - 155 b^4 + 12 b^2 + 1) a^{14} - 4 b^2 (b^{28} + 76 b^{26} - 539 b^{24} \\ & - 2118 b^{22} + 5211 b^{20} + 36360 b^{18} + 92394 b^{16} + 78414 b^{14} + 38058 b^{12} + 17992 b^{10} - 2085 b^8 \\ & - 518 b^6 + 37 b^4 + 12 b^2 + 1) a^{18} + 8 \left(b^{20} + b^{18} - 699 b^{16} + 2369 b^{14} - \frac{1037}{2} b^{12} - 1827 b^{10} \right. \\ & + \frac{2163}{2} b^8 - 159 b^6 - 27 b^4 + b^2 + 1 \left. \right) b^6 a^{22} - 1024 b^9 (b^2 + 1)^4 a^7 + 8 b^6 \left(b^{20} + b^{18} \right. \\ & - 27 b^{16} + 33 b^{14} - \frac{2189}{2} b^{12} - 3875 b^{10} - \frac{9357}{2} b^8 - 3231 b^6 - 2747 b^4 - 1055 b^2 - 127 \left. \right) \\ & a^{10} + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} - 1372 b^{18} - 9036 b^{16} - 25400 b^{14} - 54798 b^{12} - 64184 b^{10} \\ & - 27212 b^8 - 6620 b^6 - 2636 b^4 + 36 b^2 + 1) a^{12} + (-1024 b^{22} + 72 b^{20} + 692 b^{18} + 8 b^{16} \\ & - 76 b^{14} + 72 b^{12}) a^{26} + (-254 b^{28} + 840 b^{26} + 6504 b^{24} - 696 b^{22} - 25752 b^{20} - 169584 b^{18} \\ & - 181788 b^{16} - 85104 b^{14} - 50328 b^{12} - 4792 b^{10} + 360 b^8 + 72 b^6 + 2 b^4) a^{20} + (b^{32} + 8 b^{30} \\ & + 20 b^{28} + 1672 b^{26} + 1950 b^{24} - 32376 b^{22} - 157780 b^{20} - 317080 b^{18} - 279672 b^{16} \\ & - 222104 b^{14} - 125524 b^{12} - 7288 b^{10} - 3426 b^8 - 120 b^6 + 20 b^4 + 8 b^2 + 1) a^{16} + (-9 b^{24} \\ & - 72 b^{22} + 144 b^{20} - 1912 b^{18} + 142 b^{16} + 4232 b^{14} + 2192 b^{12} - 1864 b^{10} - 265 b^8) a^8 \\ & + (72 b^{20} - 76 b^{18} + 1288 b^{16} + 692 b^{14} - 696 b^{12} - 512 b^{10}) a^6 + (6 b^{20} - 32 b^{18} + 44 b^{16} \\ & - 288 b^{14} - 250 b^{12}) a^4 - 2560 \left(b^6 - \frac{38}{5} b^4 + \frac{42}{5} b^2 - \frac{11}{5} \right) b^{11} (b^2 + 1)^4 a^{21} \\ & - 1536 b^9 \left(b^8 - \frac{43}{3} b^6 + \frac{95}{3} b^4 - \frac{52}{3} b^2 + \frac{7}{3} \right) (b^2 + 1)^4 a^{19} + 512 b^7 (b^{10} + 11 b^8 - 78 b^6 \\ & + 81 b^4 - 22 b^2 + 1) (b^2 + 1)^4 a^{17} + 512 b^7 (b^{10} - 10 b^8 - 9 b^6 + 42 b^4 - 19 b^2 + 1) (b^2 \\ & + 1)^4 a^{15} - 2560 \left(b^8 - \frac{16}{5} b^6 + b^4 - \frac{3}{5} b^2 + \frac{3}{5} \right) b^7 (b^2 + 1)^4 a^{13} + 2560 \left(b^6 - \frac{6}{5} b^4 \right. \end{aligned}$$

$$+ \frac{8}{5} b^2 - 1 \Big) b^7 (b^2 + 1)^4 a^{11} + 512 b^7 (b^2 - 2) (b^2 + 1)^5 a^9 \Big) :$$

#Using the implicit function theorem, compute $\frac{da}{db} = D2$ in terms of a and b

$$D2 := - \frac{\text{diff}(r2, b)}{\text{diff}(r2, a)} :$$

#We construct the curve $D2 - D1$ where its zeros describe points where the derivative of $\left((w + z)^3 - y^3 \right) \cdot x \cdot \left(-w^2 \cdot (z^3 - y^3) + z^2 \cdot (w^3 - y^3) \right) \Big/ \left(y \cdot (w + z)^2 \cdot \left((x^3 - w^3) \cdot (z^3 - y^3) + (x^3 - z^3) \cdot (w^3 - y^3) \right) \right)$ is zero

$$f := D2 - D1 :$$

#Calculate the numerator of f

$$\tilde{f} := \text{numer}(f) :$$

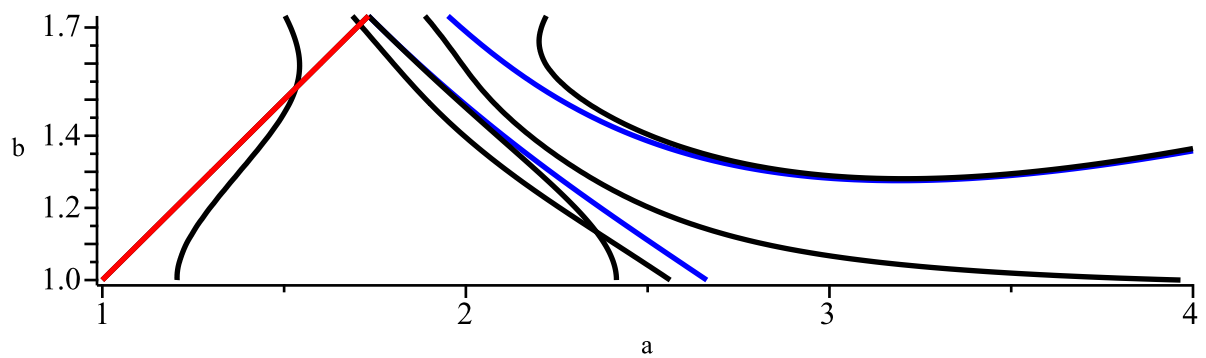
#Plot the curves $r2=0$ and $\tilde{f}=0$.

$$r := a - b :$$

with(plots) :

```
implicitplot([ r2=0,  $\tilde{f}$ =0, r=0], a = 1 .. 4, b = 1 .. (sqrt(3)),
    scaling = constrained,
    gridrefine = 2,
    thickness = 2,
    color = [blue, black, red],
    labels = ["a", "b"],
    title = "Graphic for r2=0 and  $\tilde{f}$ =0");
```

Graphic for $r_2=0$ and $\tilde{f}=0$



#Let's build the DescartesSingChanges function that counts the number of sign changes between the coefficients of a given polynomial.

```
DescartesSignChanges := proc (polynomial, variable)  
  local coeffs, nonzero_coeffs, s, i, n;  
  
  # Get the list of coefficients from highest degree to constant term  
  coeffs := [seq(coeff (polynomial, variable, i), i = degree (polynomial, variable) .. 0, -1)];  
  
  # Remove zero coefficients (they do not count in Descartes' Rule)  
  nonzero_coeffs := remove (c → c = 0, coeffs);  
  
  # Count the number of sign changes  
  n := nops (nonzero_coeffs);  
  s := 0;  
  for i from 1 to n - 1 do  
    if signum (nonzero_coeffs[i]) * signum (nonzero_coeffs[i + 1]) = -1 then  
      s := s + 1;  
    end if;  
  end do;  
  
  return s;  
end proc;
```

#Compute the resultant between $r2$ and $\tilde{f}$ with respect to a

$R1 := \text{resultant}(r2, \tilde{f}, a) :$

#Compute the factors of the resultant $R1$ and count how many there are

```

dgb := proc(f)
  if type(f, ``) then
    degree(op(1,f), b);
  else
    degree(f, b);
  end if;
end proc;

bfator := sort(
  [op(factor(R1))],
  (f, g) → dgb(f) < dgb(g)
);

```

$\text{nops}([op(factor(R1))]);$

15

(2)

#We list below the factors of the resultant $R1$, except for the factor 15 because it is too large. When necessary, we will use Sturm's algorithm to check if the factors cancel each other out within the interval $[1, \sqrt{3})$ where γ_2 is defined. For the $rb15$ factor, because of its high degree, we will use Moebius transformations as it requires less processing power.

$rb1 := bfator[1];$

$rb1 := -206294879089744981779124551495182122153863456743895446897623467852562432$ (3)

$rb2 := bfator[2];$

$rb2 := b^{1296}$ (4)

$rb3 := bfator[3];$

$rb3 := b^2 + 4b + 1$ (5)

$rb4 := bfator[4];$

$rb4 := b^2 - 4b + 1$ (6)

$rb5 := bfator[5];$

$rb5 := (b^2 + 1)^{494}$ (7)

$rb6 := bfator[6];$

$rb6 := (b^2 - 3)^5$ (8)

$rb7 := bfator[7];$

$\text{sturm}(\text{sturmseq}(rb7, b), b, 1, \sqrt{3});$ *#number of roots in $[1, \sqrt{3})$*

$rb7 := b^4 - 4b^3 + 18b^2 - 4b + 1$

$$0 \quad (9)$$

$rb8 := bfator[8];$

$$rb8 := b^4 + 4 b^3 + 18 b^2 + 4 b + 1 \quad (10)$$

$rb9 := bfator[9];$

$sturm(sturmseq(rb9, b), b, 1, \sqrt{3})$; #number of roots in $[1, \sqrt{3})$)

$$rb9 := (7 b^4 - 6 b^2 + 3)^5 \quad (11)$$

$rb10 := bfator[10];$

$sturm(sturmseq(rb10, b), b, 1, \sqrt{3})$; #number of roots in $[1, \sqrt{3})$)

$$rb10 := 47 b^6 - 195 b^4 + 141 b^2 - 65 \quad (12)$$

$rb11 := bfator[11];$

$sturm(sturmseq(rb11, b), b, 1, \sqrt{3})$; #number of roots in $[1, \sqrt{3})$)

$$rb11 := (15 b^6 - 3 b^4 + 45 b^2 - 1)^2 \quad (13)$$

$rb12 := bfator[12];$

$sturm(sturmseq(rb12, b), b, 1, \sqrt{3})$; #number of roots in $[1, \sqrt{3})$)

$$rb12 := (17 b^6 + 3 b^4 + 51 b^2 + 1)^2 \quad (14)$$

$rb13 := bfator[13];$

$sturm(sturmseq(op(1, rb13), b), b, 1, \sqrt{3})$; #number of roots in $[1, \sqrt{3})$)

$$\begin{aligned} rb13 := & (45241 b^{48} - 983136 b^{46} + 8765868 b^{44} - 53430800 b^{42} + 292102794 b^{40} \\ & - 1327146624 b^{38} + 4932365356 b^{36} - 14813913168 b^{34} + 36572797215 b^{32} \\ & - 74566977600 b^{30} + 126486144024 b^{28} - 177535429920 b^{26} + 204746625068 b^{24} \\ & - 190367984256 b^{22} + 140202896376 b^{20} - 79729549344 b^{18} + 34666181511 b^{16} \\ & - 11137697376 b^{14} + 2222925500 b^{12} - 25011408 b^{10} - 137706774 b^8 + 38273024 b^6 \\ & - 2821284 b^4 - 1082256 b^2 + 374017)^{10} \end{aligned} \quad (15)$$

$rb14 := bfator[14];$

$sturm(sturmseq(op(1, rb14), b), b, 1, \sqrt{3})$; #number of roots in $[1, \sqrt{3})$)

$$\begin{aligned} rb14 := & (201545550647774641 b^{156} + 26091219170754511998 b^{154} \\ & + 361585232998046982939 b^{152} - 130963328751372183668 b^{150} \\ & - 130388126939757400959735 b^{148} - 279967648023527012610558 b^{146} \\ & + 3620019864154595920678667 b^{144} + 160940106010701027705944424 b^{142} \\ & - 1972979746709284943883794829 b^{140} + 13927305220149044325187283410 b^{138} \\ & - 79539059828434235052427832607 b^{136} + 310481324910885734882601124524 b^{134} \end{aligned}$$

$$\begin{aligned}
& + 182598334338368271586785687811 b^{132} - 12005386098267790709457670903890 b^{130} \\
& + 92300599687783329815049589360713 b^{128} \\
& - 494978415415112481078763544152128 b^{126} \\
& + 2023145351439782210888740989477924 b^{124} \\
& - 4173537101566204561731174651183240 b^{122} \\
& - 13032733636063165568183302585300468 b^{120} \\
& + 171791709178326296999845457915208048 b^{118} \\
& - 1039795400253525481034862310322193468 b^{116} \\
& + 4829981885821823924585546414792734088 b^{114} \\
& - 17468220296985996262470337578279187572 b^{112} \\
& + 46121682190537227842779749333610917216 b^{110} \\
& - 81107980734204901315267408424455543716 b^{108} \\
& + 76388760308187455911174939910611108968 b^{106} \\
& - 12072987991564623810576421275785035212 b^{104} \\
& + 116614354222940452510620157173133006896 b^{102} \\
& - 1058590461372895421771282653974865740708 b^{100} \\
& + 3540004390856713390340444922603703221912 b^{98} \\
& - 7013659464620873378334148195447331743852 b^{96} \\
& + 8833375032775702807948649709388563447360 b^{94} \\
& - 6563805327143655203940509170540028090466 b^{92} \\
& + 3955205261570191183123912384193755367108 b^{90} \\
& - 14856122074723924124298860722562802725814 b^{88} \\
& + 55979376886560926403656740401524260235304 b^{86} \\
& - 128483772924084915191410765495527023342930 b^{84} \\
& + 205307506061458741821964537288964359204668 b^{82} \\
& - 243591856351288543568598444641772571843158 b^{80} \\
& + 220004269000606531937914985420979881281136 b^{78} \\
& - 153561585682384177912187927176499510824182 b^{76} \\
& + 90729161554414712574897887505486629206332 b^{74} \\
& - 61656051419883747450238738450383134563058 b^{72} \\
& + 58724410174258598649006211920296209780776 b^{70} \\
& - 54063291285060043601451079668741746632662 b^{68} \\
& + 35824619471723942177739204744538576767172 b^{66} \\
& - 14564401603684848684541945022041763295618 b^{64}
\end{aligned}$$

$$\begin{aligned}
& + 4732008090683601101198916412388181802560 b^{62} \\
& - 5360558687346575145805290233225679935404 b^{60} \\
& + 7256092189817071284720504677788303708824 b^{58} \\
& - 4966509761132987187123479749849535262948 b^{56} \\
& + 1536436454887031164160118447625589712944 b^{54} \\
& + 15866249334274782087646245588033249780 b^{52} \\
& + 301516206764056666239464248319853242472 b^{50} \\
& - 572304726582372003943456557444324658660 b^{48} \\
& + 343615675526010254474949876055978241376 b^{46} \\
& - 51063565384779734300946344512014119220 b^{44} \\
& - 22149057370604876964509659678447189624 b^{42} \\
& - 12357330403805598669075164837520603900 b^{40} \\
& + 20983044594507402382571162605982323056 b^{38} \\
& - 10762446129766802461316300362871686068 b^{36} \\
& + 3003570392540101865087764748741303160 b^{34} \\
& - 486560265839604073196474197116043164 b^{32} \\
& + 14506644890019352043843463903049664 b^{30} \\
& + 8323582120659525168536263620251961 b^{28} \\
& - 619860369954219114028234260172626 b^{26} - 111426996676113263095944985685901 b^{24} \\
& + 9900984422511556818979295917740 b^{22} - 49863654734008661388943836783 b^{20} \\
& + 162340076637417985627564578514 b^{18} - 13829661974246732966859355485 b^{16} \\
& - 1956213362565512717830833816 b^{14} + 216048149459477528028020827 b^{12} \\
& + 17407565191158737689970946 b^{10} - 1764554824835978005223463 b^8 \\
& - 34068413508632845696116 b^6 - 1002344030045036839125 b^4 \\
& + 347072952894808621950 b^2 - 4170153564935809983)^2
\end{aligned}$$

0

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#The factor rb15 has degree 612, so we will not show it, just compute how many zeros it has in the desired range.

rb15 := bfator[15]:

degree(rb15 , b);

612

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#From previous tests, using the Moebius transformation on the rb15 factor was more efficient when subdividing the interval [1, sqrt(3)) into two subintervals: [1, 1.4), and [1.4, sqrt(3)).

#Apply the Möbius transformation on the interval (1, 1.4), constructing Rb15_1

$$bmoebius1 := \left\{ b = \frac{1 + \left(\frac{7}{5} \right) d}{1 + d} \right\} :$$

Rb15_1 := collect(numer(expand(subs(bmoebius1, rb15))), d) :

#From here we will analyze the signal variation of all coefficients Rb15_1

DescartesSignChanges(Rb15_1, d);

1

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#Apply the Möbius transformation on the interval (1.4, sqrt(3)), constructing Rb15_2

$$bmoebius2 := \left\{ b = \frac{\frac{7}{5} + (\sqrt{3})d}{1 + d} \right\} :$$

Rb15_2 := collect(numer(expand(subs(bmoebius2, rb15))), d) :

#From here we will analyze the signal variation of all coefficients Rb15_2

DescartesSignChanges(Rb15_2, d);

1

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#Finding possible candidates for zeros of the factor rb15

fsolve(rb15=0, b, 1 .. (\sqrt{3}));

1.345726290, 1.629433656

(20)

#We now repeat the same process for the resultant of r_2 with $\tilde{f}$, in relation to b

$R_2 := \text{resultant}(r_2, \tilde{f}, b) :$

#Compute the factors of the resultant R_2 and count how many there are

```

dg(a) := proc(f)
  if type(f, '^') then
    degree(op(1,f), a);
  else
    degree(f, a);
  end if;
end proc;

afator := sort(
  [op(factor(R2))],
  (f, g) → dg(a)(f) < dg(a)(g)
) :

```

$\text{nops}([op(factor(R2))])$;

15

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#We list below the factors of the resultant R_2 , except for the factor 15 because it is too large. When necessary, we will use Sturm's algorithm to check if the factors cancel each other out within the interval $[\text{sqrt}(3), 4)$ where $\mathcal{P}$ is defined. For the ra_{15} factor, because of its high degree, we will use Moebius transformations as it requires less processing power.

$ra_1 := \text{afator}[1]$;

$ra_1 := -206294879089744981779124551495182122153863456743895446897623467852562432$ (22)

$ra_2 := \text{afator}[2]$;

$ra_2 := a^{1392}$ (23)

$ra_3 := \text{afator}[3]$;

$ra_3 := a^2 + 4a + 1$ (24)

$ra_4 := \text{afator}[4]$;

$ra_4 := a^2 - 4a + 1$ (25)

$ra_5 := \text{afator}[5]$;

$ra_5 := (a^2 + 1)^{494}$ (26)

$ra_6 := \text{afator}[6]$;

$ra_6 := (a^2 - 3)^5$ (27)

$ra_7 := \text{afator}[7]$;

$\text{sturm}(\text{sturmseq}(ra_7, a), a, \text{sqrt}(3), 4)$; #number of roots in $[\text{sqrt}(3), 4)$

$ra_7 := a^4 - 4a^3 + 18a^2 - 4a + 1$

$$0 \quad (28)$$

$ra8 := afator[8];$

$$ra8 := a^4 + 4 a^3 + 18 a^2 + 4 a + 1 \quad (29)$$

$ra9 := afator[9];$

$sturm(sturmseq(ra9, a), a, \sqrt{3}, 4); \# \text{number of roots in } [\sqrt{3}, 4)$

$$ra9 := (7 a^4 - 6 a^2 + 3)^5$$

$$0 \quad (30)$$

$ra10 := afator[10];$

$sturm(sturmseq(ra10, a), a, \sqrt{3}, 4); \# \text{number of roots in } [\sqrt{3}, 4)$

$$ra10 := 47 a^6 - 195 a^4 + 141 a^2 - 65$$

$$1 \quad (31)$$

#Finding possible candidates for zeros of the factor ra10

$fsolve(ra10=0, a, \sqrt{3} .. 4);$

$$1.839289820 \quad (32)$$

$ra11 := afator[11];$

$sturm(sturmseq(ra11, a), a, \sqrt{3}, 4); \# \text{number of roots in } [\sqrt{3}, 4)$

$$ra11 := (15 a^6 - 3 a^4 + 45 a^2 - 1)^2$$

$$0 \quad (33)$$

$ra12 := afator[12];$

$$ra12 := (17 a^6 + 3 a^4 + 51 a^2 + 1)^2 \quad (34)$$

$ra13 := afator[13];$

$sturm(sturmseq(op(1, ra13), a), a, \sqrt{3}, 4); \# \text{number of roots in } [\sqrt{3}, 4)$

$$ra13 := (175559 a^{48} - 1769112 a^{46} + 6749652 a^{44} + 9460584 a^{42} - 193871370 a^{40} \\ + 750687384 a^{38} - 111548076 a^{36} - 9218817960 a^{34} + 37541503329 a^{32} \\ - 86577381616 a^{30} + 144267501288 a^{28} - 188241754608 a^{26} + 198904792788 a^{24} \\ - 173518719696 a^{22} + 126058415880 a^{20} - 76187388496 a^{18} + 37964657145 a^{16} \\ - 15394822392 a^{14} + 5027018436 a^{12} - 1306370040 a^{10} + 287741334 a^8 - 59068744 a^6 \\ + 10970148 a^4 - 1480584 a^2 + 96383)^{10}$$

$$0 \quad (35)$$

$ra14 := afator[14];$

$sturm(sturmseq(op(1, ra14), a), a, \sqrt{3}, 4); \# \text{number of roots in } [\sqrt{3}, 4)$

$$ra14 := (44405496165211489 a^{156} + 778432081743373806 a^{154} + 4588983301097861547 a^{152} \\ + 91463528253800665452 a^{150} + 1735997934385890809049 a^{148} \\ + 11834475584313712598610 a^{146} - 251950730909385181924677 a^{144} \\ - 346879421248819184453976 a^{142} + 19687900365414339934332867 a^{140} \\ - 991610283863121627224829598 a^{138} - 2330734140754831613723067951 a^{136}$$

$$\begin{aligned}
& + 43644494250936302528980825740 a^{134} - 1185266340216425882511662940621 a^{132} \\
& + 965498050544660381854646053470 a^{130} + 51827298493210383572604009983769 a^{128} \\
& - 798953306339655826800382819027520 a^{126} \\
& + 2418561486352732770247081063125732 a^{124} \\
& + 22505999118362201449613086293175608 a^{122} \\
& - 291979589000490034255976063805058996 a^{120} \\
& + 1201823407849735701767493862937572080 a^{118} \\
& + 2293620483252666120653229957450482436 a^{116} \\
& - 54968349747308004622619101750438484280 a^{114} \\
& + 298815787919895766023209219153001367116 a^{112} \\
& - 565170501387507477526976272173989191584 a^{110} \\
& - 2853091661016888289000999525164207068516 a^{108} \\
& + 27985760058406148166512064129422041599912 a^{106} \\
& - 123829999220749404322651513548950921755404 a^{104} \\
& + 364131146300339550961244987663105308453808 a^{102} \\
& - 770693666491457720745309482695477487189988 a^{100} \\
& + 1191542222972832359072223854754521997252696 a^{98} \\
& - 1306622591090883817394234393674596223295532 a^{96} \\
& + 880308412623124256747281454141863757631552 a^{94} \\
& + 11346970395126931728531340682395108604478 a^{92} \\
& - 1161007460020240103947662927112728331749916 a^{90} \\
& + 2501060970095357501980378629424664977799466 a^{88} \\
& - 3799336966843999476217846615982967355699608 a^{86} \\
& + 4216464510482355401209665009526725091213582 a^{84} \\
& - 2878653164479224683536112576321112323318628 a^{82} \\
& + 22782768104289547570883952701155905332298 a^{80} \\
& + 2949579172284263875517603053513135217112560 a^{78} \\
& - 4515095823857240315069222923490007949934742 a^{76} \\
& + 4086074928051909692126718458679077736733980 a^{74} \\
& - 2248582796136185023260114303018516675317714 a^{72} \\
& + 93688929788265465678894977082613567396968 a^{70} \\
& + 1439483793322603463170171355171460441997002 a^{68} \\
& - 1889354214789446704832152008454964815010716 a^{66} \\
& + 1397501719614025495696613521508333409952734 a^{64}
\end{aligned}$$

$$\begin{aligned}
& - 501464125075972356709761348946349688967104 a^{62} \\
& - 268751379640310899986208819214716043596780 a^{60} \\
& + 651556703278695497184558623528782089349464 a^{58} \\
& - 671516898313972531371291330569851480619940 a^{56} \\
& + 500660534350972827100611625354864176805808 a^{54} \\
& - 301158198299676427416570645288899746788300 a^{52} \\
& + 152986091428732734077127308639814945162408 a^{50} \\
& - 67223130227783141775368744635172989695524 a^{48} \\
& + 25898918782711300837751466027572834451552 a^{46} \\
& - 8822427556502312342130422113129913421300 a^{44} \\
& + 2671586257758021895076610864274710191560 a^{42} \\
& - 720758027732669146959818597806343051580 a^{40} \\
& + 172995997550938533001785912480586741488 a^{38} \\
& - 36747319342870175142158551198807008500 a^{36} \\
& + 6830149278312126704277975923523008568 a^{34} \\
& - 1083297494058569654128894561687341660 a^{32} \\
& + 138274403647061857259270315687417280 a^{30} \\
& - 11901539844399448942914922276770711 a^{28} \\
& + 10234444613452164094595065416606 a^{26} + 239918939823066018790705974028675 a^{24} \\
& - 54754172262602794250844978956916 a^{22} + 7534724591572525692260253704673 a^{20} \\
& - 652220249958489356161152918622 a^{18} + 18381275079956913389760038355 a^{16} \\
& + 4466827715637094880030762664 a^{14} - 860863706423618339042756309 a^{12} \\
& + 80923719155728268966782482 a^{10} - 4623401410378690841280183 a^8 \\
& + 207351687482767870069356 a^6 - 13154712311724387999525 a^4 \\
& + 773931217904095649454 a^2 - 18927294346712438127)^2
\end{aligned}$$

1

(36)

#Finding possible candidates for zeros of the factor ra14

fsolve(*op*(1, *ra14*) = 0, *a*, *sqrt*(3) .. 4);

3.474957787

(37)

$$ra13 := a^4 - 4 a^3 + 18 a^2 - 4 a + 1$$

(38)

#The factor ra15 has degree 376, so we will not show it, just compute how many zeros it has in the desired range.

ra15 := *afator*[15]:

```
degree( ra15 , a);
```

612 (39)

#Apply the Möbius transformation on the interval $(\sqrt{3}, 4)$, constructing Ra15

$$amoebius := \left\{ a = \frac{\sqrt{3} + 4 \cdot c}{1 + c} \right\} :$$

```
Ra15 := collect(numer(expand(subs(amoebius, ra15))), c) :
```

#From here we will analyze the signal variation of all coefficients Ra15

```
DescartesSignChanges(Ra15, c);
```

2 (40)

#Finding possible candidates for zeros of the factor ra15

```
fsolve(ra15=0, a, sqrt(3) .. 4);
```

3.883374171, 3.888068164 (41)

#Let's rule out that the root $a=2+\sqrt{3}$ does not form a pair with any of the other candidates b

#Apply the Möbius transformation on the interval $[1, \sqrt{3})$, constructing $\tilde{r}_2$

$$moebius := \left\{ b = \frac{1 + \sqrt{3} d}{1 + d} \right\} :$$

$\tilde{r}_2 := \text{collect}(\text{numer}(\text{expand}(\text{subs}(moebius, \text{eval}(r_2, a = 2 + \sqrt{3}))))), d) :$

#From here we will analyze the signal variation of all coefficients $\tilde{r}_2$

$\text{DescartesSignChanges}(\tilde{r}_2, d);$

1 (42)

$\text{fsolve}(\text{eval}(r_2, a = 2 + \sqrt{3}) = 0, b, 1 .. \sqrt{3})$

1.316165044 (43)